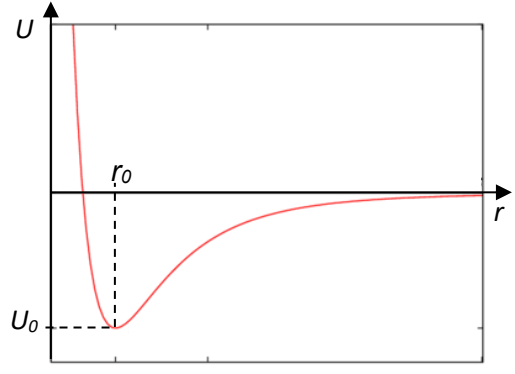
20 The simplified interaction between a pair of atoms can be represented by an equation relating the electric potential energy of the atoms,

$$U = \frac{A}{r^{12}} - \frac{B}{r^{6}}$$

where U is the electric potential energy of the atoms, r is the separation between them and A and B are constants. The variation of U with r is as shown below.



Which of the following statements is correct?

A The atoms experience no interatomic forces at the separation of r_0.

B The atoms experience the strongest forces of attraction as r decreases from r_0.

C To separate the atoms, the atoms must decrease electric potential energy by a magnitude of U_0.

D As the separation between the atoms increases from r_0, the magnitude of net force acting on the atoms increases then decreases.